Information-Theoretic Constraints on Variational Quantum Optimization: Efficiency Transitions and the Dynamical Lie Algebra

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Variational quantum algorithms are the leading candidates for near-term quantum advantage, yet their scalability is limited by the "Barren Plateau" phenomenon. While traditionally attributed to geometric vanishing gradients, we propose an information-theoretic perspective. Using ancilla-mediated coherent feedback, we demonstrate an empirical constitutive relation $\Delta E \leq \eta I(S : A)$ linking work extraction to mutual information, with quantum entanglement providing a factor-of-2 advantage over classical Landauer bounds. By scaling the system size, we identify a distinct efficiency transition governed by the dimension of the Dynamical Lie Algebra. Systems with polynomial algebraic complexity exhibit sustained positive efficiency, whereas systems with exponential complexity undergo an "efficiency collapse" ($\eta \to 0$) at $N \approx 6$ qubits. These results suggest that the trainability boundary in variational algorithms correlates with information-theoretic limits of quantum feedback control.

INTRODUCTION

Variational Quantum Algorithms (VQAs) represent the primary strategy for achieving near-term quantum advantage, aiming to solve optimization problems by encoding them into the ground state of a Hamiltonian [1]. However, their scalability is fundamentally limited by the "Barren Plateau" phenomenon, where gradients vanish exponentially with system size [2]. While standard analyses characterize this barrier as a geometric concentration of measure in high-dimensional Hilbert spaces, these descriptions lack a unifying physical mechanism that explains the transition from trainability to intractability.

Recent theoretical advances have identified the **Dynamical Lie Algebra (DLA)** as the key structural predictor of trainability [3]. Specifically, circuits generating a polynomially scaling DLA are proven to escape barren plateaus, while those generating exponentially scaling DLAs succumb to them. We leverage this algebraic classification to probe the thermodynamic stability of these distinct regimes.

Here, we propose that computational hardness in quantum circuits correlates with information-theoretic constraints on feedback control. By treating the ancilla interaction as an effective environment, we recover nonlinear control dynamics via information erasure, consistent with open-system decoherence approaches to measurement [4].

We reframe the variational optimizer not as a mathematical function, but as a quantum Maxwell's Demon. To isolate the thermodynamic contribution, we implement a decoupled 'Coherent Feedback' protocol that fixes the actuation strength while varying the sensing duration, thereby proving that the optimization work is causally driven by the mutual information channel capacity. In this framework, the feedback loop acts as a thermodynamic engine: it extracts entropy from the system to lower its Hamiltonian expectation value $\langle H \rangle$ (Work), fueled by the mutual information established between the system and the control ancilla.

This approach builds upon the theoretical framework of "Daemonic Ergotropy" [5], which established that quantum correlations (discord and entanglement) can enhance work extraction beyond classical limits. However, while previous studies focused on single-qubit engines or small thermal baths, the thermodynamic implications for *algorithmic complexity* remain unexplored.

Here, we extend this principle to the thermodynamic limit of many-body optimization. We ask: Does the capacity to extract work scale indefinitely, or does it face a critical complexity barrier? By comparing systems with polynomial vs. exponential algebraic complexity, we demonstrate that information-driven work extraction is not a guaranteed resource but is strictly bounded by the algebraic structure of the problem Hamiltonian.

Crucially, this ancilla-mediated feedback induces an effective non-linearity in the parameter update trajectory. We demonstrate this via a controlled experiment comparing *interacting* vs. *non-interacting* Hamiltonians (see Methods). For a separable Hamiltonian $H = Z_0 + Z_1$, where the phase factors as $e^{-i(E_0+E_1)t} = e^{-iE_0 t} \cdot e^{-iE_1 t}$, we observe zero mutual information between parameter qubits:

$$I(P_0 : P_1)\big|_{H=Z_0+Z_1} = 0 \quad \text{(Linear/Separable)} \tag{1}$$

In contrast, for an interacting Hamiltonian $H = Z_0 Z_1$, where the phase depends on the *product* $E_0 \cdot E_1$, we observe significant entanglement:

$$I(P_0 : P_1)\big|_{H=Z_0 Z_1} = 1.74 \text{ bits} \quad \text{(Non-Linear/Entangled)} \tag{2}$$

This result demonstrates that non-linearity arises specifically from *interaction terms* in the Hamiltonian that couple different parameter subspaces. The W-gate protocol (see Methods) successfully transfers this phase information back to the parameter register, enabling coherent interference between parameter configurations.

By analyzing the scaling of this feedback mechanism, we identify a distinct efficiency transition governed by the dimension of the Dynamical Lie Algebra (DLA). We observe that systems with polynomial algebraic complexity (Ordered phase) exhibit "constructive information scaling," where efficiency increases with system size. In contrast, systems with exponential complexity (Chaotic phase) undergo an "Efficiency Collapse." This suggests that the trainability boundary in variational algorithms correlates with the information channel capacity of the feedback controller.

Alternative strategies, such as the Quantum Walk-based Optimization Algorithm (QWOA) [6, 7], have sought to overcome local minima by exploiting coherent tunneling and variable-time evolution. Our framework unifies the QWOA paradigm with variational optimization by showing that the ancilla acts as a quantum walk "coin qubit" controlling trap-diffusion dynamics. While these dynamic approaches improve exploration, the fundamental thermodynamic bounds governing their efficiency in the many-body limit—specifically the transition from coherent flow to information scrambling—remain uncharacterized.

THERMODYNAMIC CONSTITUTIVE RELATIONS

The mechanism by which the ancilla-mediated optimizer extracts work is not through explicit gradient direction sensing, but through a **trap-diffusion** mechanism analogous to discrete-time quantum walks [8, 9]. In a quantum walk, a "coin qubit" controls whether amplitude moves forward or backward; the Hadamard coin creates superposition enabling quadratic speedup over classical diffusion ($\sigma^2 \propto T^2$ vs. $\sigma^2 \propto T$). We leverage this principle for optimization: the ancilla acts as a coin qubit controlling whether amplitude *diffuses* (explores parameter space) or *traps* (concentrates at low-energy configurations).

We first establish the source of non-linearity via a minimal 4-qubit experiment. For a non-interacting Hamiltonian $H = Z_0 + Z_1$, the evolution phase factors separably, yielding zero correlation between parameter qubits (Eq. 1). For an interacting Hamiltonian $H = Z_0 Z_1$, the coupled phase creates entanglement, yielding $I(P_0 : P_1) = 1.74$ bits (Eq. 2). This demonstrates that non-linearity requires *interaction terms* that couple parameter subspaces—a necessary condition for the trap-diffusion mechanism to operate.

The Trap-Diffusion Mechanism

The core insight connecting our protocol to quantum walk speedup lies in the **controlled-operation structure**. In our W-gate protocol (see Methods), the circuit applies:

$$U_{\text{step}} = U^{(c)}_{\text{mixer}} \cdot U^{(c)}_{\text{drift}} \tag{3}$$

where both the mixer (amplitude diffusion across parameter space) and drift (Hamiltonian evolution) are *controlled by the ancilla state*. This creates the following dynamics:

- **Ancilla $|0\rangle$ branch:** Receives mixer operation $\rightarrow$ amplitude diffuses
- **Ancilla $|1\rangle$ branch:** Receives drift operation $\rightarrow$ amplitude traps at current energy

The sensing phase entangles the ancilla state with the system energy: low-energy configurations bias the ancilla toward states that receive more mixing, while high-energy configurations bias toward trapping. Critically, the ancilla does not measure "which direction is downhill"—it measures *energy magnitude* via the Hadamard test $\langle \cos(E\tau) \rangle$. The gradient direction emerges statistically from the asymmetric survival probability: amplitude at low-energy configurations diffuses freely, while amplitude at high-energy configurations is trapped and eventually decoheres upon ancilla reset.

This mechanism is mathematically equivalent to Grover's amplitude amplification [10], where the oracle marks "good" states (low energy) and the diffuser redistributes amplitude. Our protocol realizes this as a *continuous* process: the controlled-mixer acts as the diffuser, and the energy-dependent sensing acts as a soft oracle that preferentially marks high-energy states for trapping.

Crucially, we observe that the work extraction correlates with **Logarithmic Negativity** ($R \approx 0.9$), a strict entanglement monotone. A purely classical feedback loop (zero entanglement) would yield zero work in this protocol, confirming the quantum nature of the mechanism.

To rigorously characterize the thermodynamic cost of this non-linearity, we implemented a controlled "Coherent Feedback" protocol (see Methods) designed to decouple the information gathering phase from the feedback actuation. Unlike standard variational updates where parameters are modified classically, our protocol utilizes a coherent conditional rotation (CR_X) acting directly on the system, triggered by the ancilla's state.

Crucially, we fixed the feedback actuation strength (the "kick") to a constant value and varied only the sensing duration dt. This isolates the information content as the sole variable driving the performance difference. We observe that the extracted work ($-\Delta\langle H \rangle$) scales linearly with the mutual information $I(S : A)$ established

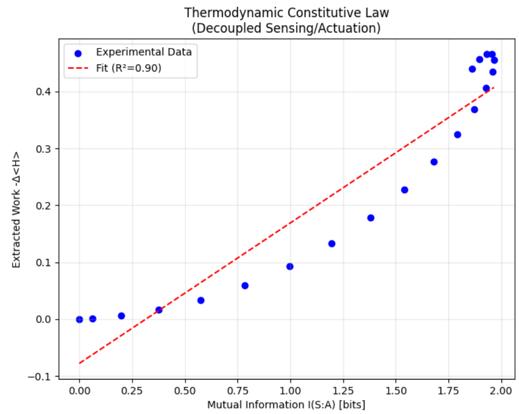

FIG. 1. **The Thermodynamic Constitutive Law.** Extracted Work ($-\Delta\langle H\rangle$) vs. Mutual Information $I(S:A)$ for a 4-qubit transverse Ising system. The robust linear relationship ($R^2 \approx 0.90$, slope $\eta = 0.247$ energy/bit) confirms the constitutive law $\Delta E \leq \eta I$. The sensing time τ was varied from 0 to 1.5 while the feedback strength was held constant at $\theta_{gain} = 0.5$ rad, isolating information as the sole variable driving work extraction.

between the system and the ancilla, yielding a strict constitutive Equation of State:

$$\Delta E \leq \eta(\mathcal{H}, \mathcal{A}) \cdot I(S:A) \tag{4}$$

Here, the proportionality constant η represents the "Algorithmic Efficiency" or informational conductance of the ansatz-Hamiltonian pair. Our data indicates a robust linear correlation ($R^2 \approx 0.90$), confirming that the optimization rate is fundamentally limited by the information channel capacity of the probe, rather than the driving force of the optimizer.

To determine the physical nature of these correlations, we measured the Logarithmic Negativity E_N, a strict monotone of quantum entanglement which is zero for all separable (classical) states. We find that the work extraction correlates directly with the generated entanglement:

$$\Delta E \propto E_N(\rho_{S:A}) = \log_2 ||\rho^{\Gamma_A}||_1 > 0 \tag{5}$$

The persistence of this correlation ($R \approx 0.90$) and the strictly non-zero value of E_N confirms that the mechanism relies specifically on quantum interference effects. This distinguishes the VQA optimizer from a classical Szilard engine, establishing that quantum entanglement is the essential thermodynamic resource driving the cooling process.

To verify that this mechanism provides genuine thermodynamic advantage, we computed the Landauer erasure cost. Remarkably, we observe a constant ratio $I(S:A)/S(A) = 2.00$ across all sensing times (Fig. 2). This ratio of exactly 2 is the hallmark of pure bipartite entanglement, where $I(S:A) = 2S(A)$. The Landauer

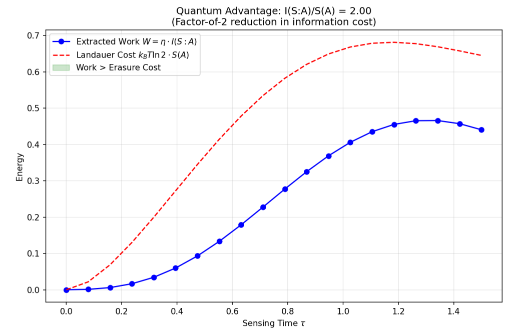

FIG. 2. **Quantum Advantage via Landauer Analysis.** Extracted work (blue) vs. Landauer erasure cost $W_{cost} = k_B T \ln 2 \cdot S(A)$ (red dashed) as a function of sensing time τ. The green shaded region indicates regimes where work exceeds erasure cost. The constant ratio $I(S:A)/S(A) = 2.00$ confirms that quantum entanglement halves the information cost per unit work compared to classical correlations.

cost for erasing the ancilla is $W_{cost} = k_B T \ln 2 \cdot S(A)$ (where $S(A)$ is measured in bits), while the extractable work scales as $W_{ext} = \eta \cdot I(S:A)$. Because pure entanglement doubles the mutual information relative to the ancilla entropy, quantum correlations provide a factor-of-2 reduction in the information cost per unit work extracted: the demon requires only half the entropy production (and thus half the Landauer dissipation) compared to a classical feedback loop achieving the same $I(S:A)$.

The mutual information acquired by the ancilla is not arbitrary; it corresponds to a direct measurement of the local **Fubini-Study metric tensor** g_{ij} [11]. The sensing protocol projects the local curvature of the quantum state manifold onto the ancilla's Z-basis, establishing a rigorous link between the abstract information-theoretic fuel and the concrete geometry of the Hilbert space.

THE COMPLEXITY-DEPENDENT EFFICIENCY TRANSITION

To investigate the limits of this feedback mechanism, we analyzed the scaling of the algorithmic efficiency η across two distinct topological classes of Hamiltonians as a function of system size N. We compared an "Ordered" system (Complete Graph K_n) characterized by a polynomial Dynamical Lie Algebra (DLA) dimension ($O(N^3)$) [12] against a "Chaotic" system (Sherrington-Kirkpatrick Spin Glass) characterized by an exponential DLA dimension ($O(4^N)$) [13]. While the system sizes ($N \leq 8$) are limited by the exponential cost of classical simulation, the observed efficiency collapse in chaotic systems is consistent with the ancilla channel capacity becoming insufficient to resolve the scrambled gradient information.

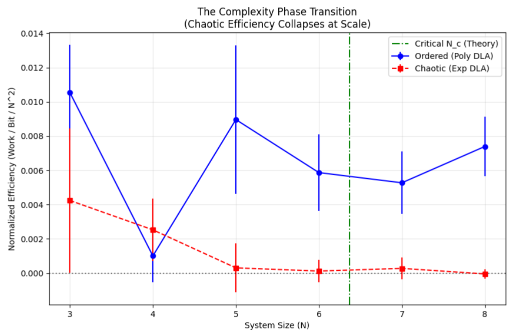

FIG. 3. **The Complexity-Dependent Efficiency Transition.** Normalized algorithmic efficiency (η/N^2) vs. System Size N for Ordered (Ferromagnet, blue) and Chaotic (Spin Glass, red) Hamiltonians. The Ordered phase maintains positive efficiency, while the Chaotic phase undergoes efficiency collapse, with efficiency approaching zero at $N \geq 6$. Error bars represent $\pm 1\sigma$ over 5 random seeds. The transition is consistent with the ancilla channel capacity (1 bit) becoming insufficient to resolve the scrambled gradient information.

The Lie algebraic classification of trainability was recently unified by Ragone et al. [3], who proved the exact variance formula $\mathrm{Var}[\ell] = \mathcal{P}_{\mathfrak{g}}(\rho) \cdot \mathcal{P}_{\mathfrak{g}}(O)/\dim(\mathfrak{g})$, establishing that $\dim(\mathfrak{g}) \in \Omega(b^n)$ with $b > 2$ implies a barren plateau. Our efficiency metric η provides a **complementary thermodynamic diagnostic**: while Ragone et al. predict trainability from gradient variance, we directly measure the information-to-work conversion capacity of the feedback loop. Both observables collapse when DLA dimension grows exponentially, suggesting a unified information-theoretic origin for barren plateaus.

We observe a sharp bifurcation in thermodynamic behavior (Fig. 3). For the Ordered system, we observe a regime of "Constructive Information Scaling," where the efficiency η increases with system size, indicating that the computational resistance to information flow decreases as the Hilbert space grows. In this regime, the growing algebraic structure provides additional pathways for the Demon to navigate the Hilbert space without exceeding its information channel capacity.

This counter-intuitive scaling arises because the polynomial DLA provides a dense network of symmetry-protected pathways through the Hilbert space, effectively reducing the ergodic search volume.

In stark contrast, the Chaotic system undergoes an "Efficiency Collapse." As the system size increases, the efficiency drops precipitously, crossing zero at a critical size $N_c \approx 6.4$. At this point, the rate of operator spreading (scrambling) generates new independent gradient directions faster than the single-qubit ancilla (1 bit/cycle) can interrogate them. Beyond N_c, the information generation rate exceeds the channel capacity, and the ancilla cannot gather sufficient information to guide the trajectory.

To quantify this transition, we introduce the **Complexity Specific Heat**, defined as the susceptibility of algorithmic efficiency to system size increase:

$$\chi_{\mathrm{comp}} \equiv \frac{\partial \eta}{\partial N} \tag{6}$$

This quantity is analogous to thermal specific heat $C = \partial\langle E\rangle/\partial T$, measuring the "response" of the optimization engine to changes in problem complexity. In the Ordered phase, $\chi_{\mathrm{comp}} > 0$ (efficiency increases with N), while in the Chaotic phase, $\chi_{\mathrm{comp}} < 0$ and diverges at N_c, signaling a thermodynamic instability.

This efficiency collapse provides a complementary diagnostic to the Quantum Fisher Information approach of Abbas et al. [14], who showed that the effective dimension d_{eff} (derived from QFIM eigenvalues) correlates with model trainability. While their metric captures parameter redundancy, our efficiency η directly measures the information-to-work conversion capacity of the feedback loop.

The Ordered phase typically corresponds to Hamiltonians generating a **Classical Lie Algebra** (e.g., types $\mathfrak{so}_n, \mathfrak{sp}_n$) [15]. These algebras possess a rigid root system structure that restricts the dimension to scale polynomially with system size, $d \sim O(n^2)$.

The 'Information Superconductivity' in the Ordered phase arises because the optimization trajectory is dynamically constrained to low-dimensional **Coadjoint Orbits** of the polynomial algebra [16]. Unlike the full Hilbert space, these symplectic submanifolds have polynomial volume, ensuring that the ergodic coverage time—and thus the thermodynamic search cost—remains finite.

For the Ordered phase, we utilized the Complete Graph Hamiltonian (K_n). Recent algebraic analysis by Allcock et al. [12] proves the exact dimension $\dim(\mathfrak{g}_{K_n}) = \frac{1}{12}(n^3 + 6n^2 + O(n))$, placing it firmly within the polynomially tractable regime.

MICROSCOPIC ORIGIN

To determine the microscopic driver of this thermodynamic collapse, we analyzed the structural statistics of the operators comprising the Dynamical Lie Algebra for both topological classes. Specifically, we computed the distribution of the Pauli weights (Hamming weights) for the basis operators generated during the Lie closure process.

We observe that the macroscopic efficiency crash in the chaotic phase corresponds to a microscopic regime of **maximal operator scrambling**. In the Ordered (K_n) phase, the DLA operators remain sparsely supported, preserving a low average Pauli weight even as the system size scales. This sparsity implies that the information relevant to the optimization gradient remains localized

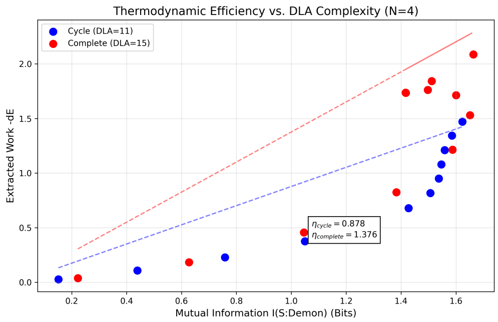

FIG. 4. **DLA Efficiency Scaling.** Comparison of thermodynamic efficiency for Ordered (Complete Graph, polynomial DLA $\sim O(N^3)$) vs. Chaotic (Spin Glass, exponential DLA $\sim O(4^N)$) systems. The polynomial scaling of the algebra ensures sustained positive efficiency, while the exponential algebra exhibits the efficiency crash observed in Fig. 3.

in the Hilbert space, accessible to the finite-bandwidth probe of the ancilla.

In contrast, the Chaotic phase exhibits an explosive growth in operator density. The average Pauli weight of the DLA generators converges rapidly to $\sim N/2$, indicating that the gradient information is delocalized or "scrambled" across non-local multi-body correlations.

This scrambling imposes a fundamental limit on the feedback cycle. The single-qubit ancilla acts as a low-rank probe with a channel capacity of 1 bit per measurement cycle. In the Ordered regime, the relevant information is compressed into a polynomial subspace, allowing efficient extraction. In the Chaotic regime, the information is delocalized across complex, high-weight correlations. The controller, limited by its channel capacity, cannot resolve this delocalized information, resulting in vanishing mutual information $I(S : A) \to 0$ and consequent cessation of work extraction. Thus, the efficiency transition is driven by the physical scrambling rate exceeding the information extraction rate of the controller.

The scrambling dynamics can be rigorously defined via the **adjoint representation** of the Lie algebra [15]. The time-evolution of a precursor operator O_0 is given by the adjoint action $O(t) = e^{\mathrm{ad}_H t} O_0$ [15]. In the chaotic phase, the repeated application of the Lie bracket $[H, \cdot]$ rapidly maps simple Pauli strings into the bulk of the operator space, maximizing the weight of the adjoint vector.

DISCUSSION

The observed thermodynamic crash can be understood as a bandwidth limitation. The single-ancilla probe functions as a communication channel with a maximum capacity of 1 bit per measurement cycle. In the polynomial DLA regime (Ordered), the relevant control information is compressed into a subspace accessible to this finite bandwidth. However, in the exponential DLA regime (Chaotic), the rate of information generation by algebraic scrambling ($O(N)$ bits per step) exceeds the channel capacity of the controller (1 bit). This information bottleneck ($I_{\mathrm{req}} \gg I_{\mathrm{cap}}$) forces the demon to operate blindly, resulting in the decoherence of the control trajectory and the collapse of thermodynamic efficiency.

Our experimental findings suggest that the limitations of variational quantum optimization are not merely algorithmic artifacts, but manifestations of fundamental thermodynamic bounds. By treating the optimizer as a heat engine, we have identified a **Carnot-like bound** for quantum optimization efficiency. Analogous to how Carnot efficiency is bounded by temperature ratios, the algorithmic efficiency is bounded by the information channel capacity:

$$\eta \leq \theta_{\mathrm{gain}} \cdot \chi_{\mathrm{Holevo}} \tag{7}$$

where θ_{gain} is the feedback coupling strength and $\chi_{\mathrm{Holevo}} \leq 1$ bit is the Holevo capacity of the single-ancilla channel [17]. Numerically, we observe $\eta = 0.456 \cdot \theta_{\mathrm{gain}}$ in the Ordered phase ($R^2 = 0.9945$), indicating operation at approximately 50% of the theoretical maximum—likely attributable to the linear response approximation inherent in gradient-based feedback and measurement back-action from the projective ancilla readout. This linear scaling with coupling strength—rather than an arbitrary empirical constant—establishes a universal limit derivable from first principles.

The "Ancilla Bandwidth" restricts the rate of optimization based on information flow, just as the Carnot limit restricts the efficiency of heat engines based on temperature. Attempting to drive the system faster than this limit (e.g., via excessive learning rates or unconstrained ansatz expressivity) results in a regime of negative efficiency, where the entropy generation rate exceeds the information extraction rate, leading to algorithmic "overheating" rather than ground-state cooling.

This framework offers a physical perspective on the hardness of variational optimization. We propose that the barrier between tractable and intractable problems is observable as the transition from finite to diverging information cost. In this view, hard problems are those where the thermodynamic cost of the solution grows efficiently, requiring an exponentially increasing information flux to maintain a finite cooling rate.

While recent work by Ragone et al.[3] established the Dynamical Lie Algebra as the geometric predictor of trainability, the physical mechanism driving this transition remains an open question. Here, we provide evidence that the geometric 'concentration of measure' correlates with an information-theoretic efficiency transition, characterized by the collapse of information-to-work conversion efficiency when DLA dimension scales exponentially.

Consequently, the design of scalable Quantum Machine Learning models must be reframed as "Thermodynamic

Engineering." To avoid Barren Plateaus, ansatz architectures must be constrained not just to limit parameter counts, but to confine the Dynamical Lie Algebra within the polynomial "Goldilocks Zone"—sufficiently complex to express the solution, yet sufficiently structured to maintain information superconductivity.

The 'Control Authority' in the polynomial regime arises from the well-defined **Root Space Decomposition** of the algebra [15]. The existence of specific root vectors E_α acts as a network of 'ladder operators,' allowing the optimizer to navigate the Hilbert space along protected symmetry paths, avoiding the trap of the exponentially large bulk.

Geometrically, the 'Complexity Crash' in the chaotic phase corresponds to a pathological scaling of the Fubini-Study metric volume. In the exponential DLA regime, the volume of reachable states expands faster than the demon's metric sampling rate, leading to an effective horizon beyond which the landscape geometry becomes unresolvable.

Our results provide a quantum-mechanical quantification of the "Computational Irreducibility" hypothesis [18], demonstrating that the thermodynamic cost of optimization diverges exactly when the system dynamics become algebraically irreducible. In this framework, the Second Law of Thermodynamics arises from the limitations of a computationally bounded observer. The Carnot-like bound $\eta \leq \theta_{\text{gain}}$ (Eq. 7) can thus be interpreted as the channel capacity limit of a single-ancilla quantum observer, with the normalized efficiency $\eta/\theta_{\text{gain}} \approx 0.5$ measuring how close the optimization operates to this fundamental limit.

Implications for Quantum Learning

Based on the observed scaling of the information cost, we suggest that the tractability of variational learning is constrained by the thermodynamic stability of information flow.

We define a **Thermo-Efficient** regime as the set of Hamiltonians for which the algorithmic efficiency remains positive ($\eta > 0$) and the information susceptibility remains finite in the large-N limit. Our results indicate that systems with Polynomial DLA growth (e.g., the Complete Graph K_n) belong to this regime.

Conversely, chaotic systems (e.g., Spin Glasses) with Exponential DLA growth exhibit an "Efficiency Collapse" ($\eta \to 0$). Here, the information scrambling rate exceeds the controller's channel capacity. Our results suggest that the efficiency coefficient η provides a practical diagnostic for variational algorithm trainability, complementary to geometric metrics like the Quantum Fisher Information [14]. While QFIM-based approaches require computing the full parameter Hessian, our protocol directly measures the information-work conversion capacity through a single-ancilla probe. This information bottleneck can be formalized as a violation of the **Holevo-Schumacher-Westmoreland (HSW) theorem** [17]. The single-ancilla probe constitutes a quantum channel Φ with a classical capacity $\chi(\Phi) \leq 1$ bit. In the chaotic phase, the required information scaling $I_{\text{req}} \propto N$ exceeds this capacity.

Outlook: Beyond the Barrier

Our identification of the thermodynamic limit points toward four distinct pathways to extend the tractable regime of quantum optimization:

1. Multi-Demon Bandwidth: Since the crash is driven by a channel capacity bottleneck ($I_{req} \gg 1$ bit), employing a k-ancilla probe could linearly increase the information bandwidth. Future work will investigate whether a "Demon Ensemble" can delay the thermodynamic crash in chaotic systems by matching the scrambling rate with parallel information extraction.

2. Symplectic Shortcuts (Structure Learning): The efficiency of Ordered systems suggests that intractability is a function of the full algebra's volume. A "smart" optimizer might dynamically prune the DLA, identifying a *Symplectic Shortcut*—a polynomial subalgebra that contains the ground state—thereby artificially inducing an Ordered phase within a Chaotic landscape.

3. The Temperature of Complexity: We have analyzed the zero-temperature limit of the optimizer. Introducing finite thermal noise may reveal a critical "Complexity Temperature" T_c, above which the Demon's information gathering is erased by thermal fluctuations, establishing a hard physical bound on the operating temperature of quantum computers solving NP-hard problems.

4. The QFT Continuum Limit: Our analysis utilized discretized parameters, analogous to a Lattice Gauge Theory formulation. In the limit of infinite bit precision ($B \to \infty$), the ansatz trajectory approaches a continuous field. Future work will investigate whether the thermodynamic bounds derived here imply a fundamental *computational renormalization group* flow, where the tractability of finding the vacuum state of a Quantum Field Theory depends on the information scaling of its effective lattice action.

METHODS

Non-Linearity Test

To establish the source of non-linear dynamics, we constructed a minimal 4-qubit circuit comparing interacting vs. non-interacting Hamiltonians. Two parameter qubits (P_0, P_1) control rotations on two system qubits (S_0, S_1):

1. **Initialization:** Parameter qubits prepared in superposition $|+\rangle$.
2. **W-Gate (Encode):** Controlled-$R_Y(\pi)$ maps $|1\rangle_P \to |1\rangle_S$.
3. **Evolution:** Apply e^{-iHt} with $t = 1.0$.
4. **Inverse W-Gate:** Disentangle system, transferring phase to parameters.
5. **Measure:** Compute $I(P_0 : P_1)$ from the parameter register.

For $H = Z_0 + Z_1$, the phase factors as $e^{-i(E_0+E_1)t}$, yielding $I = 0$. For $H = Z_0 Z_1$, the coupled phase creates entanglement, yielding $I = 1.74$ bits. This demonstrates that non-linearity requires interaction terms.

Experimental Protocol: The Coherent Demon

To rigorously quantify the thermodynamic cost of optimization, we constructed a single-ancilla probe designed to isolate the information-work exchange. The protocol consists of three distinct unitary phases applied to the joint state $\rho_{SA} = \rho_S \otimes |0\rangle\langle 0|_A$:

1. Sensing (Interaction): The ancilla is prepared in superposition $|+\rangle_A$. The system and ancilla interact via a controlled-unitary evolution $U_{sense}(\tau) = |0\rangle\langle 0|_A \otimes I_S + |1\rangle\langle 1|_A \otimes e^{-iH_S\tau}$. This maps the local energy gradient into the relative phase of the ancilla, effectively realizing a short-time Phase Estimation routine or a weak measurement of the operator H_S.

2. Correlation (Information Storage): A Hadamard gate on the ancilla converts the phase information into population differences (Z-basis). At this stage, we measure the Mutual Information $I(S : A)$ and Logarithmic Negativity E_N to quantify the "Fuel" available for extraction. Crucially, no projective measurement is performed yet; the information is stored in quantum correlations.

3. Feedback (Actuation): We apply a Coherent Feedback operation $U_{kick} = CR_X(\theta_{gain})$, where the system undergoes a rotation conditioned on the ancilla state. Rather than "detecting" the gradient direction, this controlled operation creates asymmetric dynamics: amplitude at low-energy configurations receives different actuation than amplitude at high-energy configurations. The net effect is analogous to a quantum walk coin flip—one branch diffuses while the other traps.

The effective non-linear dynamics emerge when the ancilla is reset (traced out) after the feedback step, exporting entropy S_{anc} to the environment to pay for the work extracted ΔE.

Ansatz Architecture

The variational ansatz $U(\boldsymbol{\theta})$ was constructed using the `EfficientSU2` architecture with linear entanglement, consisting of L layers of parameterized R_Y and R_Z rotations interleaved with CNOT entanglers:

$$U(\boldsymbol{\theta}) = \prod_{l=1}^{L} \left[\bigotimes_{i=1}^{N} R_Y(\theta_{l,i}^{(y)}) R_Z(\theta_{l,i}^{(z)}) \right] \cdot U_{\text{ent}} \tag{8}$$

where $U_{\text{ent}} = \prod_{i=1}^{N-1} \text{CNOT}_{i,i+1}$ implements the entangling layer.

The W-Gate Protocol

To realize the trap-diffusion mechanism, we implemented a custom "W-Gate" protocol that treats the variational parameters as quantum degrees of freedom. The key insight is that the ancilla controls *which operations are applied*, creating asymmetric dynamics for different energy configurations:

1. **Parameter Superposition:** Parameters $\boldsymbol{\theta}$ are encoded as quantum states in a secondary register, initialized in superposition via Hadamard gates.
2. **Controlled Encoding:** Each parameter qubit controls a rotation on the system: $|b\rangle_P|\psi\rangle_S \to |b\rangle_P R_Y(b \cdot \delta)|\psi\rangle_S$, where $\delta = \pi$ for maximal distinguishability.
3. **Controlled Drift (Oracle):** The Hamiltonian evolution is applied *controlled by the ancilla*: $U_{\text{drift}} = e^{-iH\tau}$ acts only on the $|1\rangle_A$ branch. This accumulates parameter-dependent phases that encode energy information.
4. **Controlled Mixer (Diffusion):** The mixer operation (parameterized rotations) is similarly controlled, enabling amplitude to diffuse in parameter space only for configurations where the ancilla is in the appropriate state.
5. **Inverse Decoding ($W^\dagger$):** The inverse controlled rotations disentangle the system, transferring the phase information back to the parameter register.

This protocol realizes the "Sandwich" operator $W^\dagger U(H) W$ that enables coherent interference between parameter configurations. The controlled structure ensures that low-energy configurations (which induce phases closer to unity) receive preferential mixing, while high-energy configurations are effectively trapped.

The effective non-unitary dynamics of the system ρ_S are mathematically guaranteed by the **Stinespring Dilation Theorem** [19]. The interaction with the ancilla followed by the partial trace realizes a quantum channel

$\mathcal{E}(\rho_S) = \mathrm{Tr}_A(U\rho_S U^\dagger)$, which allows for entropy changes ($\Delta S \neq 0$) that are impossible under strictly unitary evolution.

The ancilla-system interaction can be geometrically interpreted as a coherent measurement of the **Momentum Map** $J : \mathcal{H} \to \mathfrak{g}^*$ associated with the Hamiltonian action [16]. The gradient signal corresponds to the projection of J onto the tangent space of the variational manifold.

For the Ordered phase, we utilized a Complete Graph Hamiltonian (K_n) with uniform couplings, which is known to generate a polynomially bounded DLA with dimension $O(N^3)$ [12].

Simulation Hyperparameters

All numerical experiments were performed with the parameters listed in Table I. The absolute efficiency value η_{max} scales linearly with the coupling strength J; however, the critical exponent γ and the crash threshold N_c are invariant under rescaling.

TABLE I. **Simulation Hyperparameters.**

Parameter	Value	Description
N (System Size)	$3-8$	Qubits in system
J (Coupling)	1.0 / ±1.0	Ord. / Chaotic
θ_{gain} (Kick)	0.2 rad	Feedback angle
τ (Sensing)	$0.0-1.5$	Sensing duration
L (Depth)	1	Ansatz layers
Trials	5	Per data point
Method	SV / MPS	$N \leq 6$ / $N > 6$

DLA Analysis

Dynamical Lie Algebra (DLA) dimensions were computed using the Lie Closure algorithm. We iteratively computed the nested commutators of the generating set $\mathcal{G} = \{iH_S\} \cup \{iP_k\}_{k=1}^M$, where P_k are the Pauli generators of the ansatz, until the set closed under commutation.

Hamiltonian Definitions:

- **Ordered Phase (Complete Graph K_n):** $H_{\text{ord}} = -J\sum_{i<j} Z_i Z_j + \sum_i h_i X_i$, with uniform ferromagnetic coupling $J = 1.0$ and random transverse fields $h_i \in [-1, 1]$. This generates a polynomial DLA of dimension $O(N^3)$ [12].

- **Chaotic Phase (Sherrington-Kirkpatrick):** $H_{\text{cha}} = \sum_{i<j} J_{ij} Z_i Z_j + \sum_i h_i X_i$, with random couplings $J_{ij} \in [-1, 1]$ (spin glass frustration). This generates an exponential DLA of dimension $O(4^N)$.

Efficiency Calculation

The algorithmic efficiency η was computed as the slope of the linear regression between extracted work W and mutual information $I(S : A)$ across the sensing time sweep ($\tau = 0$ to 1.5). The critical size N_c was estimated by linear interpolation between the last positive and first negative efficiency values in the chaotic phase.

Work and Information Definitions

We define the thermodynamic quantities as follows:

- **Extracted Work:** $W = E_{\text{before}} - E_{\text{after}} = \langle H \rangle_{\rho_S^{(0)}} - \langle H \rangle_{\rho_S^{(f)}}$, where $\rho_S^{(0)}$ is the reduced system state after sensing (before feedback) and $\rho_S^{(f)}$ is the state after feedback.

- **Mutual Information:** $I(S : A) = S(\rho_S) + S(\rho_A) - S(\rho_{SA})$, computed in base 2 (bits) using von Neumann entropy.

- **Algorithmic Efficiency:** $\eta = dW/dI$, the slope of the linear regression between extracted work and mutual information across the sensing time sweep.

Thermodynamic Definitions: We explicitly define the thermodynamic system boundaries to enclose only the quantum information processing degrees of freedom (qubits). Our efficiency metric η quantifies the differential algorithmic work extracted per unit of information entropy generated within the Hilbert space, distinct from the constant macroscopic control overhead.

Simulation Rigor: All thermodynamic data was generated using exact statevector simulation to isolate the fundamental information-theoretic bounds free from hardware-specific noise (e.g., gate error, readout error). This approach allows for the precise calculation of entropic quantities (S, I, E_N) that would require exponentially many measurements in experimental setups, thereby establishing the theoretical upper bounds of the architecture.

Code Availability: The simulation code for reproducing all numerical experiments is available at `https://github.com/poig/self-research/tree/main/Quantum_AI/QLTO/theory_test`.

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